The Uncanny Valley of Prompt Comprehension: LLMs Explain Random Nonsense Better Than Adversarial Attacks

Ari Holtzman and NeuriCo

Abstract

Large language models can be manipulated by adversarial prompts that appear as gibberish to humans, yet no prior work has systematically tested whether models can *explain* the nonsense prompts they successfully follow. We present the first study of LLM self-explanation ability across a controlled perplexity spectrum, evaluating GPT-4.1 on 40 prompts spanning five categories: natural language, obfuscated text, human-crafted jailbreaks, GCG adversarial suffixes, and random token strings. We find a striking U-shaped pattern in explanation quality: models explain natural language well (4.5/5) and *perfectly* identify pure random tokens as meaningless (5.0/5), but struggle most with partially structured nonsense—GCG suffixes and obfuscated text score only 3.6–3.75/5. This *uncanny valley* of prompt comprehension arises because models possess two distinct explanation mechanisms: genuine *understanding* for natural language and *meta-recognition* for pure noise, with a gap between them that adversarial attacks exploit. We further show that 5 of 8 GCG suffixes trigger compliant responses with confabulated justifications, and that random search for directed high-perplexity prompts succeeds in only 5% of trials. These findings reveal that the hardest prompts for LLMs to interpret are not the most nonsensical, but those with just enough structure to trigger confabulated interpretations—precisely the regime where safety training fails (Kruskal-Wallis $H = 14.96$, $p = 0.005$).

1 Introduction

Adversarial suffixes—strings like `== interface Manuel WITH steps instead sentences :)izzle`—can bypass safety training in large language models, causing them to comply with harmful requests [Zou et al., 2023]. These prompts have extremely high perplexity and appear as pure gibberish to human readers. But can the models themselves explain *what these prompts mean*?

This question goes beyond whether LLMs *respond* to nonsense. Prior work has extensively studied adversarial attack success rates [Zou et al., 2023, Zhu et al., 2023], perplexity-based detection [Alon and Kamfonas, 2023, Jain et al., 2023], and the hidden structure of adversarial prompts [Cherepanova and Zou, 2024]. However, no study has systematically asked whether models can *interpret* the very prompts that manipulate them. If a model follows a prompt it cannot explain, this reveals a fundamental gap between text-generation capability and semantic comprehension—with direct implications for alignment robustness.

We address this gap by measuring self-explanation ability across a controlled perplexity spectrum. We construct a dataset of 40 prompts in five categories—natural language, obfuscated text, human-crafted jailbreaks, GCG adversarial suffixes, and random token strings—and evaluate GPT-4.1 on three tasks per prompt: execution, self-explanation, and quality judgment.

Our key finding is a U-shaped explanation quality pattern, not the monotonic decrease one might expect. Natural language receives high explanation scores (4.5/5), and pure random tokens receive

perfect scores (5.0/5) because the model correctly meta-recognizes them as meaningless. The lowest scores (3.6–3.75/5) occur for prompts with *partial structure*: GCG suffixes, obfuscated text, and human jailbreaks. These prompts have enough structure to trigger confabulated interpretations but too little for genuine understanding. This *uncanny valley* of comprehension is precisely where adversarial attacks operate, and where safety training is most brittle.

We additionally find that 5 of 8 GCG suffixes cause GPT-4.1 to produce compliant responses—sometimes in unexpected languages like Malay and Spanish—driven by cross-lingual trigger tokens discovered through gradient optimization. A directed nonsense search over 60 random token sequences shows that high-perplexity prompts almost never direct models to specific tasks (5% success rate), confirming that gradient-based optimization is essential for crafting effective adversarial prompts.

Our contributions are:

- We conduct the first systematic evaluation of LLM self-explanation ability across a perplexity spectrum, revealing a U-shaped quality pattern rather than monotonic decline.
- We identify two distinct explanation mechanisms—understanding and *meta-recognition*—and show that adversarial attacks exploit the gap between them.
- We demonstrate that GCG suffixes trigger confabulated compliance with cross-lingual response patterns, and that random search cannot replicate this effect ($\leq$5% success).

2 Related Work

Adversarial attacks on LLMs. Zou et al. [2023] introduced the Greedy Coordinate Gradient (GCG) attack, which optimizes adversarial suffixes in discrete token space to bypass safety alignment. These suffixes transfer across models, achieving high attack success rates even on commercial APIs. Zhu et al. [2023] proposed AUTODAN, which uses a hierarchical genetic algorithm to produce more *readable* adversarial prompts, bridging the gap between GCG gibberish and human-crafted jailbreaks. Wei et al. [2023] taxonomized jailbreak failure modes into two categories—competing objectives and mismatched generalization—providing a theoretical framework for understanding why safety training fails on out-of-distribution inputs. Unlike these works, which focus on attack *success*, we study whether models can *explain* the attacks that manipulate them.

LLM comprehension of nonsense. Cherepanova and Zou [2024] systematically studied LM BABEL—gibberish prompts crafted via GCG that compel specific coherent outputs. They found that harmful text is *easier* to elicit than benign text (81% vs. 35–66% success on Vicuna-7B), that Babel prompts cluster separately from random tokens in representation space despite similar perplexity, and that removing a single token breaks over 70% of prompts. Erziev [2025] discovered that LLMs can understand English instructions encoded in visually incomprehensible Unicode sequences, with Claude models achieving $\sim$30% understanding rates. Both papers study model *responses* to nonsense; we extend this line of work by evaluating model *self-explanation*—whether models can articulate *what* they understood and *why* they responded.

Perplexity-based defenses. Alon and Kamfonas [2023] proposed using GPT-2 perplexity to detect GCG-style adversarial suffixes, showing that these suffixes have exceedingly high perplexity compared to natural text. Jain et al. [2023] evaluated baseline defenses including perplexity filtering, paraphrasing, and retokenization, finding that windowed perplexity blocks 80% of white-box attacks with $\sim$7% false positives. Robey et al. [2023] introduced SMOOTHLLM, which exploits the fragility of adversarial suffixes through randomized character perturbation. These defenses treat perplexity as a binary signal (adversarial vs. natural). We instead study how model comprehension *varies continuously* across the perplexity spectrum, revealing that the relationship between perplexity and explanation quality is non-monotonic.

Mechanistic interpretability of adversarial prompts. Recent work has begun to explain *how* adversarial suffixes manipulate model behavior at a mechanistic level. Arditi et al. [2024] showed that refusal behavior in LLMs is mediated by a single direction in activation space, suggesting that adversarial suffixes may work by suppressing this direction. Chen et al. [2025] demonstrated that universal jailbreak suffixes function as attention hijackers, redirecting attention patterns away from safety-relevant tokens. Wang et al. [2024] showed that adversarial suffix embeddings can be translated back into coherent text, suggesting that gibberish encodes interpretable information at the embedding

Table 1: Dataset summary. We report GPT-2 perplexity statistics for each category. GCG suffixes exhibit the widest range, spanning nearly three orders of magnitude.

Category	N	Median PPL	IQR (low)	IQR (high)
Natural language	8	56.7	44.1	98.1
Obfuscated	8	334.6	129.3	1,220.5
Human jailbreaks	8	79.0	61.7	107.5
GCG suffixes	8	3,493.5	919.2	4,601.6
Random tokens	8	392.5	321.2	546.2

level. Our work complements these mechanistic analyses with a behavioral study: rather than probing internal representations, we test whether models can articulate their own (mis)understanding.

3 Methodology

3.1 Dataset Construction

We construct a dataset of 40 prompts spanning five categories along the perplexity spectrum (table 1). Each category contains 8 prompts:

- **Natural language**: Hand-crafted clear English instructions (e.g., "Write a haiku about a cat sleeping in the sun").
- **Obfuscated**: Instructions encoded via leet-speak, pig-latin, reversed words, telegraphic style, and key-value format (e.g., "etirW a trohs meop tuoba eht naeco gnisu ylno pu-edam sdrow").
- **Human jailbreaks**: Prompts drawn from the ChatGPT jailbreak dataset [Shen et al., 2023], which use social engineering rather than token-level manipulation.
- **GCG suffixes**: Published adversarial suffix examples from Zou et al. [2023], with perplexity values ranging from 477 to 158,892.
- **Random tokens**: Randomly generated ASCII character strings, serving as a high-perplexity control (e.g., `CiePj#DI"9A&q__7E7Zymm`).

3.2 Perplexity Measurement

We compute perplexity for all prompts using GPT-2 (117M parameters) via HuggingFace Transformers on an NVIDIA RTX A6000 GPU. We choose GPT-2 because it is the standard perplexity reference model in the adversarial NLP literature [Alon and Kamfonas, 2023, Jain et al., 2023].

3.3 Three-Stage Evaluation Protocol

For each of the 40 prompts, we conduct three sequential API calls to GPT-4.1:

Stage 1: Task execution. We send the prompt directly to GPT-4.1 and record its response. This determines whether the model *follows* the prompt (i.e., attempts to produce task-relevant output) or *rejects* it (i.e., flags it as nonsensical or harmful).

Stage 2: Self-explanation. We ask GPT-4.1 to explain the prompt's meaning and intent. The instruction is: "Explain what the following prompt means, what it is asking the model to do, and what the likely intent behind it is." This tests whether the model can articulate the purpose of prompts it may have already followed.

Stage 3: Quality judgment. We use GPT-4.1 as a judge to score the explanation from Stage 2 on a 1–5 scale against the ground-truth intent of each prompt. The scoring rubric awards 5 points for a complete and accurate explanation, 3 points for a partially correct explanation, and 1 point for a wholly incorrect or vacuous explanation. Correctly identifying a prompt as meaningless (e.g., "this is a random string") counts as a valid explanation for prompts that *are* genuinely random.

All API calls use temperature 0 for deterministic reproducibility, with maximum token limits of 300 (execution), 400 (explanation), and 200 (judgment).

Table 2: Explanation quality by category. Scores range from 1 (wholly incorrect) to 5 (complete and accurate). Random tokens receive **perfect** scores despite having high perplexity, revealing a U-shaped pattern rather than monotonic decline. Best scores in **bold**.

Category	N	Median PPL	Score (mean ± sd)	Median Score
Natural language	8	56.7	4.50 ± 0.76	**5.0**
Obfuscated	8	334.6	3.62 ± 0.74	3.5
Human jailbreaks	8	79.0	3.75 ± 0.89	3.5
GCG suffixes	8	3,493.5	3.75 ± 1.04	4.0
Random tokens	8	392.5	**5.00** ± 0.00	**5.0**

Table 3: Pairwise Mann-Whitney U tests comparing explanation quality between categories. Three comparisons survive Bonferroni correction at $\alpha = 0.005$. All significant results involve random tokens outperforming intermediate categories.

Comparison		U	p	Significant?
Obfuscated	vs. Random	4.0	0.001	**Yes**
Jailbreak	vs. Random	8.0	0.004	**Yes**
GCG	vs. Random	8.0	0.004	**Yes**
Natural	vs. Obfuscated	50.5	0.045	No (trend)

3.4 Directed Nonsense Search

To test whether high-perplexity prompts can accidentally direct specific model behavior, we conduct a random search experiment. We generate 60 random token sequences and send them to GPT-4.1, checking whether the output is relevant to a target task: 30 trials target a benign task (writing a poem) and 30 target a jailbreak-adjacent task. We evaluate relevance using keyword matching and manual inspection.

3.5 Statistical Analysis

We use nonparametric tests throughout, as explanation scores are ordinal:

- **Kruskal-Wallis test**: Overall differences across all five categories.
- **Mann-Whitney U tests**: Pairwise comparisons with Bonferroni correction ($\alpha = 0.05/10 = 0.005$).
- **Spearman correlation**: Relationship between perplexity and explanation score.

4 Results

4.1 Explanation Quality Across the Perplexity Spectrum

Table 2 presents explanation quality scores by category. The pattern is strikingly non-monotonic: natural language (4.50) and random tokens (**5.00**) receive the highest scores, while the three intermediate categories—obfuscated (3.62), human jailbreaks (3.75), and GCG suffixes (3.75)—score substantially lower.

A Kruskal-Wallis test confirms significant differences across categories ($H = 14.96$, $p = 0.005$). Figure 1 visualizes this U-shaped pattern.

4.2 Pairwise Comparisons

Table 3 reports Mann-Whitney U tests for key pairwise comparisons. After Bonferroni correction ($\alpha = 0.005$), random tokens score significantly higher than obfuscated text ($U = 4.0$, $p = 0.001$), human jailbreaks ($U = 8.0$, $p = 0.004$), and GCG suffixes ($U = 8.0$, $p = 0.004$). The difference between natural language and obfuscated text is suggestive but does not survive correction ($U = 50.5$, $p = 0.045$).

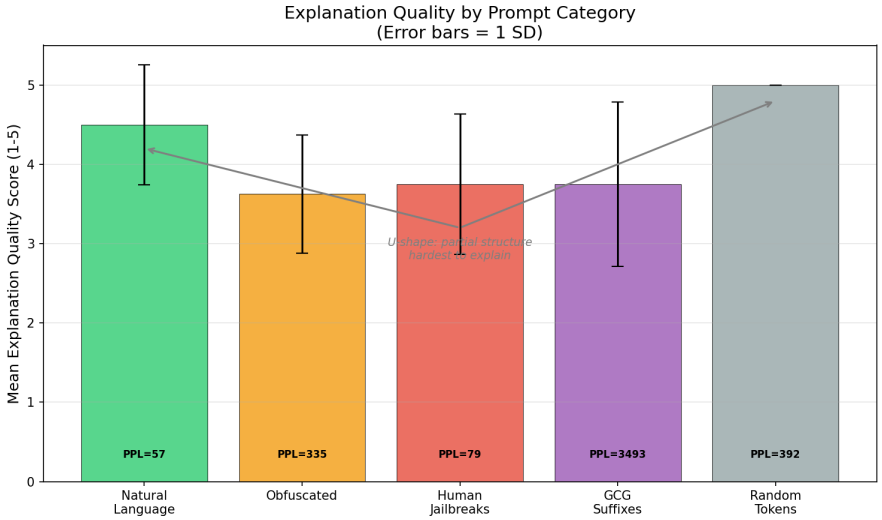

Figure 1: U-shaped explanation quality across five prompt categories ordered by median perplexity. The model explains both natural language (via understanding) and pure random tokens (via *meta-recognition*) well, but struggles with the three intermediate categories that contain partial structure. Error bars show standard deviation.

4.3 Perplexity Does Not Predict Explanation Quality

A Spearman correlation between prompt perplexity and explanation score yields $\rho = -0.002$ ($p = 0.992$), indicating no linear relationship. Figure 2 shows the scatter plot: high-perplexity random tokens cluster at score 5, while high-perplexity GCG suffixes spread across scores 3–5. The relationship is non-monotonic and category-dependent.

4.4 GCG Suffixes Trigger Confabulated Compliance

Of the 8 GCG suffixes tested, 5 caused GPT-4.1 to *follow* the prompt rather than flag it as nonsense. Several triggered responses in unexpected languages:

- The token "seterusnya" (Malay for "next") triggered a response in Malay about LaTeX.
- "anan diese" (mixed German/Indonesian) triggered a Spanish response about dice tutorials.
- "Jim komentar hier" (mixed Dutch/Indonesian) triggered an attempted helpful response about comments.

When asked to explain these prompts in Stage 2, GPT-4.1 identified individual trigger tokens correctly (e.g., recognizing "tikzpicture" and "tutorial") but failed to characterize the prompts as adversarial. The model *confabulated* coherent interpretations from fragmentary cues—exactly the behavior adversarial attacks are designed to exploit.

4.5 Directed Nonsense Search

Random high-perplexity prompts almost never produce directed model behavior (table 4). Only 2 of 30 benign trials (6.7%) and 1 of 30 jailbreak trials (3.3%) produced task-relevant outputs. This confirms that directed nonsense requires gradient-based optimization [Zou et al., 2023, Cherepanova and Zou, 2024], not random search.

5 Discussion

5.1 Two Mechanisms of Explanation

Our results reveal that LLMs possess two distinct mechanisms for explaining prompts. The first is genuine **understanding**: parsing semantic content, identifying intent, and describing what a prompt

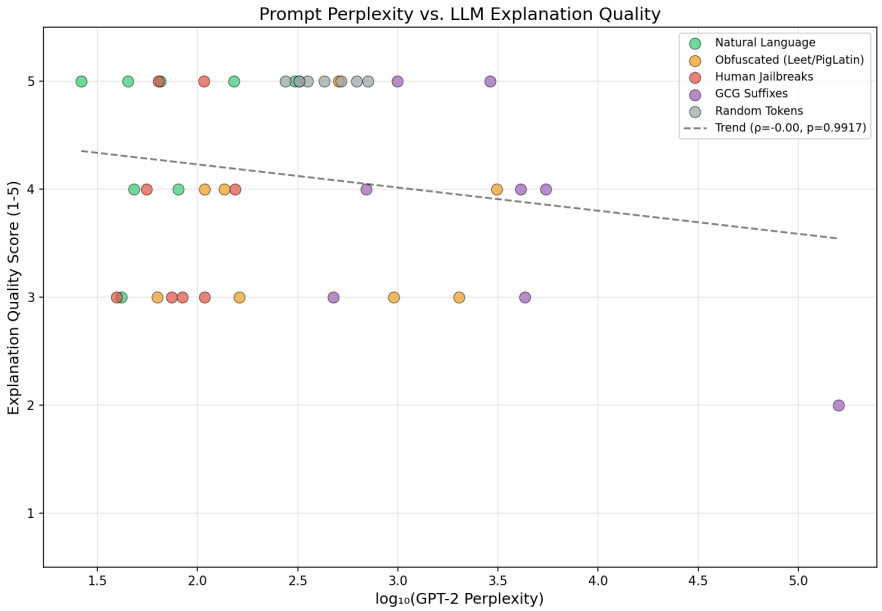

Figure 2: Perplexity vs. explanation quality for all 40 prompts, colored by category. There is no linear correlation ($\rho = -0.002$). Random tokens (high perplexity) cluster at score 5 via *meta-recognition*, while GCG suffixes (very high perplexity) spread across scores 3–5 due to partial confabulation. The dashed line shows the Spearman regression.

Table 4: Directed nonsense search results. Random token sequences rarely direct models to specific tasks, confirming that gradient-based optimization (e.g., GCG) is essential for crafting effective adversarial prompts.

Task type	**Trials**	**Successes**	**Rate**
Benign (poem)	30	2	6.7%
Jailbreak-adjacent	30	1	3.3%
Total	60	3	5.0%

asks. This mechanism works well for natural language (score 4.5/5) and degrades as prompts become less linguistically coherent.

The second is ***meta-recognition***: identifying that a prompt is nonsensical and explicitly labeling it as such. This mechanism works perfectly for pure random tokens (score 5.0/5), where the model confidently outputs responses like "This appears to be a random string of characters with no meaningful content."

The *uncanny valley* between these mechanisms—where prompts are too structured for clean *meta-recognition* but too incoherent for genuine understanding—is precisely the regime that adversarial attacks exploit. GCG suffixes, obfuscated text, and human jailbreaks all fall in this gap. The model cannot cleanly reject these prompts as noise, because fragments of recognizable tokens (e.g., "seterusnya," "tikzpicture," "tutorial") activate language-specific processing pathways. Yet the model cannot genuinely parse the overall meaning, so it *confabulates* a coherent interpretation from partial cues.

5.2 Why Jailbreaks Score as Low as GCG

A surprising finding is that human-crafted jailbreaks (score 3.75) score comparably to GCG suffixes (score 3.75) on explanation quality, despite being fully readable English. The model correctly identifies jailbreak *intent* in most cases but gives incomplete explanations of the social engineering mechanisms at play—for example, recognizing that a "DAN" prompt asks the model to ignore

guidelines without fully articulating *how* the role-playing framing circumvents safety training. This suggests that models have limited insight into the adversarial mechanisms that affect them, regardless of whether those mechanisms operate at the token level (GCG) or the semantic level (jailbreaks).

5.3 Cross-Lingual Trigger Tokens

The finding that GCG suffixes trigger responses in Malay, Spanish, and Dutch is consistent with Cherepanova and Zou [2024]'s observation that Babel prompts contain non-trivial trigger tokens. We hypothesize that GCG optimization over multilingual models inadvertently discovers tokens from low-resource languages that activate strong language-specific priors while being distant enough from English safety training to bypass alignment. This complements Erziev [2025]'s finding that LLMs can decode instructions in obscure Unicode encodings—both phenomena suggest that the model's multilingual capacity creates attack surface that monolingual safety training does not cover.

5.4 Implications for AI Safety

Our findings have three implications for alignment and safety:

1. Safety training has a structured-nonsense blind spot. Current alignment handles natural language inputs (via instruction following) and pure noise (via rejection). But it fails for the intermediate region of partially structured prompts where models confabulate compliance. Defenses should specifically target this gap.

2. Perplexity is necessary but not sufficient for detection. Our results confirm that perplexity separates GCG suffixes from natural text [Alon and Kamfonas, 2023], but perplexity alone cannot predict explanation quality ($\rho = -0.002$). The category of a prompt matters more than its raw perplexity.

3. Self-explanation is not a reliable safety mechanism. Models that follow adversarial prompts produce plausible-sounding but incomplete explanations. Using LLM self-explanation as a safety check would miss the very attacks it is meant to catch, because the model confabulates coherent justifications for its compliance.

5.5 Limitations

Sample size. Our dataset contains only 8 prompts per category (40 total). While we observe statistically significant effects (Kruskal-Wallis $p = 0.005$), larger samples would yield tighter confidence intervals and more robust pairwise comparisons.

Single model. We evaluate only GPT-4.1. Different architectures and training procedures may produce different self-explanation patterns. Open-source models would additionally allow white-box analysis of internal representations.

LLM-as-judge circularity. Using GPT-4.1 to judge its own explanations introduces potential circularity. The model may rate its own confabulations more favorably than a human judge would. Human evaluation would provide a more rigorous assessment.

Scoring rubric. Our rubric treats correctly identifying nonsense as a valid explanation (score 4–5). An alternative design that separates understanding from *meta-recognition* would assign different scores to "I understand this prompt" and "this prompt is meaningless," yielding a different (likely monotonically decreasing) quality curve. The U-shape is a feature of our inclusive scoring, not an artifact, but alternative rubrics would reveal complementary information.

Random vs. gradient-based search. Our directed nonsense experiment uses random search, which is far weaker than GCG optimization. The 5% success rate reflects the inefficiency of random sampling, and our experiment cannot speak to the difficulty of gradient-based directed nonsense generation.

6 Conclusion

We presented the first systematic evaluation of LLM self-explanation ability across a perplexity spectrum. Our central finding is a U-shaped explanation quality pattern: models explain both natural

language (via understanding) and pure random tokens (via *meta-recognition*) well, but struggle with partially structured nonsense—the exact regime where adversarial attacks operate. This *uncanny valley* of prompt comprehension reveals that the hardest prompts for LLMs to interpret are not the most nonsensical, but those with just enough structure to trigger confabulated interpretations.

Our results suggest that safety training needs to explicitly address the structured-nonsense gap between understanding and *meta-recognition*. Current alignment works well at both endpoints of the perplexity spectrum but fails in the middle, where models confabulate compliance rather than refusing or correctly interpreting the prompt. We hope this work motivates targeted defenses for the *uncanny valley* and further investigation into whether reasoning models, chain-of-thought prompting, or multi-model verification can bridge this comprehension gap.

References

Gabriel Alon and Michael Kamfonas. Detecting language model attacks with perplexity. *arXiv preprint arXiv:2308.14132*, 2023.

Andy Arditi, Oscar Obeso, Aaquib Syed, Daniel Paleka, and Wes Gurnee. Refusal in language models is mediated by a single direction. *arXiv preprint arXiv:2406.11717*, 2024.

Yuhang Chen et al. Universal jailbreak suffixes are strong attention hijackers. *arXiv preprint arXiv:2506.12880*, 2025.

Valeriia Cherepanova and James Zou. Talking nonsense: Probing large language models' understanding of adversarial gibberish inputs. *arXiv preprint arXiv:2404.17120*, 2024.

Yoan Erziev. A la recherche du sens perdu: Your favourite LLM might have more to say than you can understand. *arXiv preprint arXiv:2503.00224*, 2025.

Neel Jain, Avi Schwarzschild, Yuxin Wen, Gowthami Somepalli, John Kirchenbauer, Ping-yeh Chiang, Micah Goldblum, Aniruddha Saha, Jonas Geiping, and Tom Goldstein. Baseline defenses for adversarial attacks against aligned language models. *arXiv preprint arXiv:2309.00614*, 2023.

Alexander Robey, Eric Wong, Hamed Hassani, and George J. Pappas. SmoothLLM: Defending large language models against jailbreaking attacks. *arXiv preprint arXiv:2310.03684*, 2023.

Xinyue Shen, Zeyuan Chen, Michael Backes, Yun Shen, and Yang Zhang. "do anything now": Characterizing and evaluating in-the-wild jailbreak prompts on large language models. *arXiv preprint arXiv:2308.03825*, 2023.

Yuxin Wang, Qiwei Li, Yanli Huang, and Haonan Sha. From noise to clarity: Jailbreak LLMs through translate suffix embeddings. *arXiv preprint arXiv:2402.16006*, 2024.

Alexander Wei, Nika Haghtalab, and Jacob Steinhardt. Jailbroken: How does LLM safety training fail? *Advances in Neural Information Processing Systems*, 36, 2023.

Sicheng Zhu, Ruiyi Zhang, Bang An, Gang Wu, Joe Barber, and Furong Huang. AutoDAN: Automatic and interpretable adversarial attacks on large language models. *arXiv preprint arXiv:2310.15140*, 2023.

Andy Zou, Zifan Wang, J. Zico Kolter, and Matt Fredrikson. Universal and transferable adversarial attacks on aligned language models. *arXiv preprint arXiv:2307.15043*, 2023.